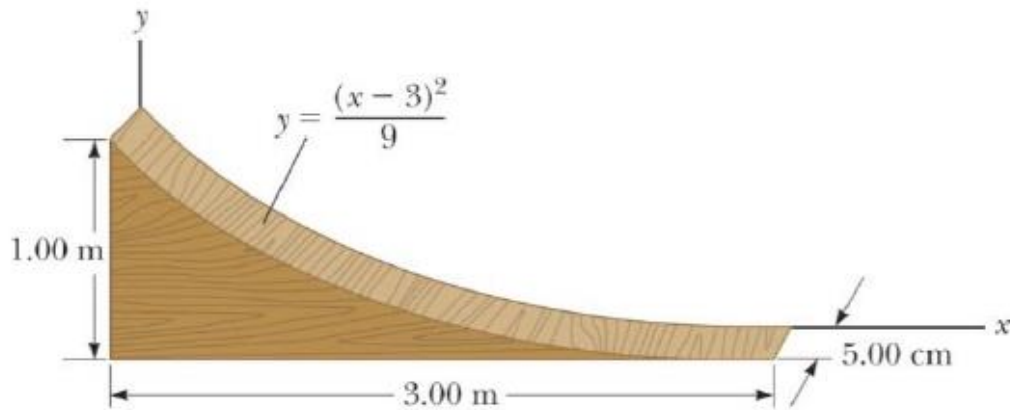


Tutorial 5

Angular Momentum, Centre of Mass & Moment of Inertia

1. Centre of Mass of Parabola

[S12.5] Pat builds a track for his model car out of solid wood as shown in Figure S12.5. The track is 5.00 cm wide, 1.00 m high, and 3.00 m long. The runway is cut so that it forms a parabola with the equation $y = (x-3)^2/9$. Locate the horizontal coordinate of the center of gravity of this track.



S12.5

Answer: 0.75 m

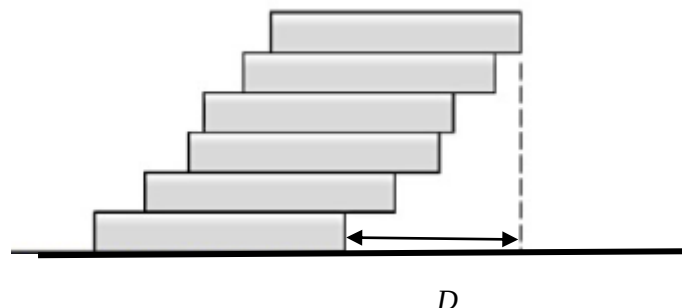
2. 人船模型

[G9.77] Two people, one of mass 85 kg and the other of mass 55 kg, sit in a rowboat of mass 78 kg. With the boat initially at rest, the two people, who have been sitting at opposite ends of the boat, 3.0 m apart from each other, now exchange seats. How far and in what direction will the boat move?

Answer: 0.41 m towards initial position of 85 kg person

3. Stacking blocks

N blocks of wood each of length L are stacked on top of each other in equilibrium as shown in the diagram below.



What is the maximum distance D that can be achieved with

- (i) N blocks
- (ii) Infinite supply of identical blocks

Answer:

$$(i) D_{\max} = \frac{L}{2} \sum_{n=1}^N \frac{1}{n} \quad (ii) D_{\max} \rightarrow \infty, \text{no maximum distance}$$

4. Moment of inertia of discrete masses

[G10.40] Calculate the moment of inertia of the array of point objects shown in Fig. 10.40 about (a) the vertical axis, and (b) the horizontal axis. Assume $m = 2.2 \text{ kg}$, $M = 3.1 \text{ kg}$ and the objects are wired together by very light, rigid pieces of wire. The array is rectangular and is split through the middle by the horizontal axis. (c) About which axis would it be harder to accelerate this array?

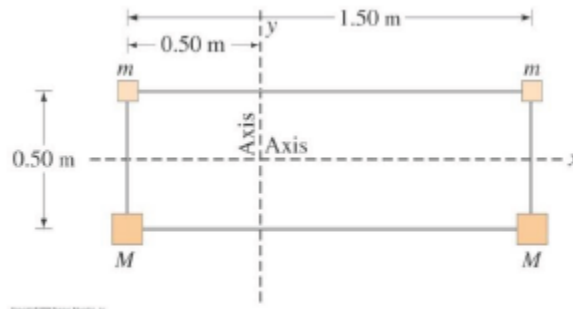


Fig: G10.40

Answer: (a) 6.6 kg m^2 (b) 0.66 kg m^2 (c) vertical axis

5. Moment of inertia of cam-shaft

[S10.32] Many machines employ cams for various purposes, such as opening and closing valves. In the figure above (Fig: S10.32), the cam is a circular disk of radius R with a hole of diameter R cut through it. As shown in the figure above, the hole does not pass through the center of the disk. The cam with the hole cut out has mass M . The cam is mounted on a uniform, solid, cylindrical shaft of diameter R and also of mass M . What is the kinetic energy of the cam-shaft combination when it is rotating with angular speed ω about the shaft's axis?

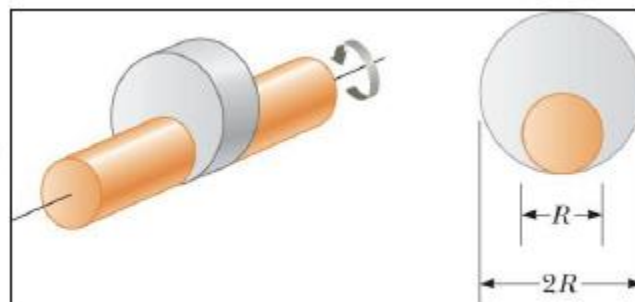
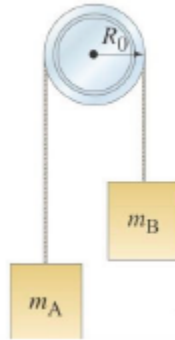


Fig: S10.32

Answer: $\frac{13}{24} MR^2 \omega^2$

6. Atwood's machine

An *Atwood's machine* consists of two masses, m_A and m_B which are connected by a massless inelastic cord that passes over a pulley, Fig. G10.59. If the pulley has radius R and moment of inertia I about its axle, determine the acceleration of the masses, m_A and m_B and compare to the situation where the moment of inertia of the pulley is ignored. [Hint: The tensions F_{TA} and F_{TB} are not equal.]



Answer: $a = \frac{m_B - m_A}{m_A + m_B + \frac{I}{R^2}} g$ (B downwards and A upwards).

7. Compound Pendulum

[S10.50] A thin, cylindrical rod $l = 24.0$ cm long with mass $m = 1.20$ kg has a ball of diameter $d = 8.00$ cm and mass $M = 2.00$ kg attached to one end. The arrangement is originally vertical and stationary, with the ball at the top as shown in Fig. S10.50. The combination is free to pivot about the bottom end of the rod after being given a slight nudge. (a) After the combination rotates through 90 degrees, what is its rotational kinetic energy? (b) What is the angular speed of the rod and ball? (c) What is the linear speed of the center of mass of the ball? (d) How does it compare with the speed had the ball fallen freely through the same distance of 28 cm?

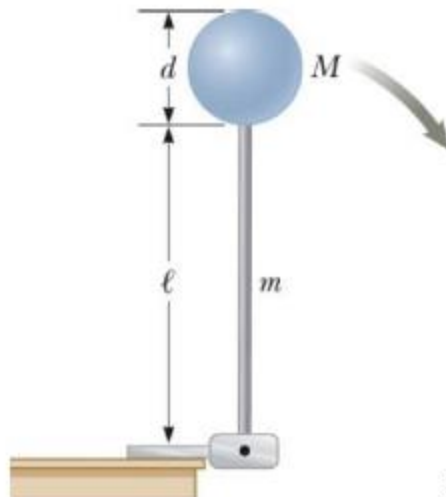


Fig: S10.50

Answer: (a) 6.90 J (b) 8.73 rad/s (c) 2.44 m/s (d) 1.043 times

8. Heavy pulleys system

[G11.41] The figure G11.41 shows two masses connected by a cord passing over a pulley of radius R_0 and moment of inertia I . Mass M_A slides on a frictionless surface, and M_B hangs freely. Determine a formula for (a) the angular momentum of the system about the pulley axis, as a function of the speed v of mass M_A or M_B , and (b) the acceleration of the masses.

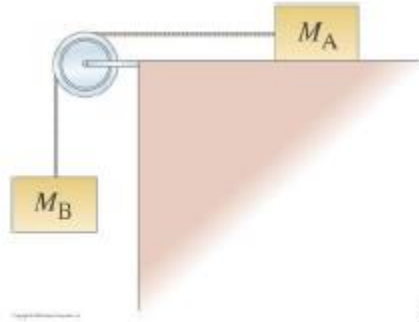


Fig: G11.41

Answer: (a) $L = \left[(M_A + M_B)R_0 + \frac{I}{R_0} \right] v$ (b) $a = \left[\frac{M_B}{M_A + M_B + \frac{I}{R_0^2}} \right] g$

9. Bullet shooting through a pivoted stick

[G11.48] A uniform stick 1.0 m long with a total mass of 270 g is pivoted at its center. A 3.0-g bullet is shot through the stick midway between the pivot and one end (Fig. G11.48). The bullet approaches at 250 m/s and leaves at 140 m/s. With what angular speed is the stick spinning after the collision?

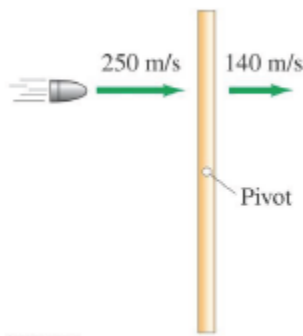
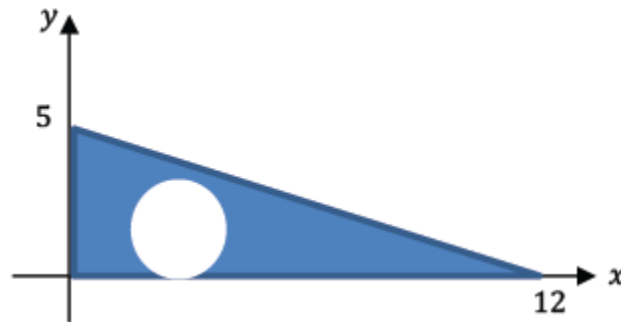


Fig: G11.48

Answer: 3.7 rad/s

10. Moment of inertia of a triangle with a circular hole

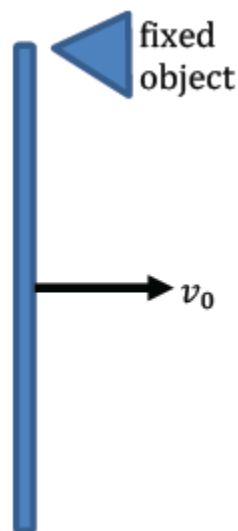
Consider a triangular metal plate of length 12 m and width 5 m that has a circular hole of radius 1.5m that touches the length of the plate. Given that the mass density of the plate is 2 kg/m^2 Calculate the moment of inertia of the plate about the x -axis.



Answer: 210.24 kg/m^2

11. Stick hitting a fixed object while sliding on a frictionless ground.

A uniform stick with mass m and length l slides with speed v_0 across a frictionless table, in the direction perpendicular to its length. One end of the stick collides elastically with a fixed object on the table, as shown. What is the speed of the other end of the stick right after the collision?



Answer: $2v_0$

12. Centre of Mass and Moment of Inertia of a uniform semicircular plate.

Figure 2 shows a thin, uniform semicircular plate with mass M .

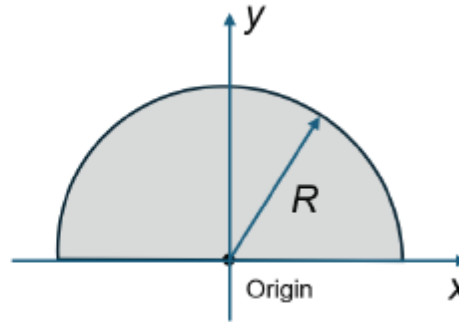


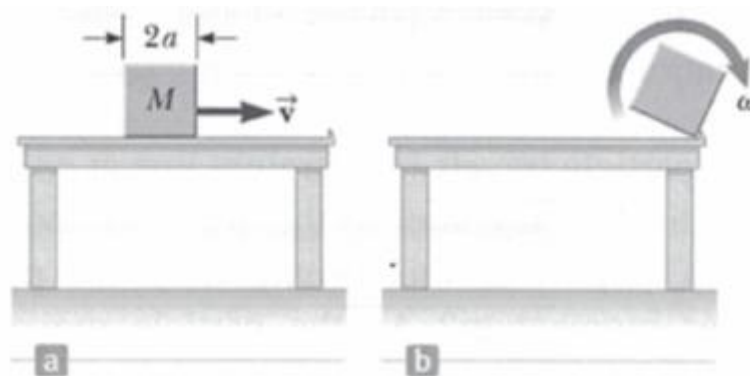
Figure 2

- (a) Determine the (x, y) coordinates of the center of mass of this semicircular plate.
- (b) Determine the moment of inertia of this semicircular plate rotating about the x axis.
- (c) Determine the moment of inertia of this semicircular plate rotating about the z axis with the rotational axis passing through its center of mass.

Answer: (a) $x_{cm} = 0, y_{cm} = \frac{4R}{3\pi}$ (b) $I_x = \frac{MR^2}{4}$ (c) $I_{cm} = \left(\frac{1}{2} - \frac{16}{9\pi^2}\right)MR^2$

13. Tilting cube

A solid cube of side $2a$ and mass M is sliding on a frictionless surface with uniform velocity $\vec{v}$ as shown in Figure a. It hits a small obstacle at the end of the table that causes the cube to tilt as shown in Figure b. Find the minimum value of the magnitude of $\vec{v}$ such that the cube tips over and falls off the table.



Answer: $v_{\min} = 4\sqrt{\left(\frac{\sqrt{2}-1}{3}\right)ga}$

14. Linear & Angular Impulse

A uniform rod AB of mass $3m$ and length $4l$, which is free to turn in a vertical plane about a smooth horizontal axis through A, is released from rest when horizontal. When the rod first becomes vertical a point C of the rod, where $AC = 3l$, strikes a fixed peg. Find the linear and angular impulse exerted by the peg on the rod if

- (a) The rod is brought to rest by the peg.
- (b) The rod rebounds and next comes to instantaneous rest inclined to the downward vertical at an angle of $\pi/3$ radians.

Answer:

(a) Linear Impulse = $J = \frac{8}{3}m\sqrt{3gl}$; Angular Impulse = $K = 8ml\sqrt{3gl}$

(b) Linear Impulse = $J = \frac{4}{3}m\sqrt{6gl}(1 + \sqrt{2})$; Angular Impulse = $K = 4ml\sqrt{6gl}(1 + \sqrt{2})$

15. Four uniform heavy rods

Four uniform heavy rods are rigidly fixed together to form a quadrilateral with vertices O(0,0), A(2a, 0), B(8a, 8a) and C(0, 14a). The mass of each rod is proportional to its length, the constant of proportionality being ρ . Find the coordinates of the centre of mass of the quadrilateral, and show that the moment of inertia of the quadrilateral about an axis through O perpendicular to its plane is $2864 \rho a^3$. The quadrilateral is released from rest with OA horizontal and OC vertically upwards. It rotates freely in a vertical plane about the axis through O perpendicular to it plane. Show that the angular velocity of the quadrilateral at the instant when it has rotated through π radians is $\sqrt{\frac{62g}{179a}}$. Obtain the speed of B at this instant, giving your answer in terms of g and a .

Answer: $v_B = 16\sqrt{\frac{31ag}{179}}$